Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of greenfield product delivery across healthcare, fintech, civic-tech, EdTech, and consumer SaaS. Deep **TypeScript** and **Node.js** on the backend (NestJS, REST APIs, Mongo and Postgres) paired with **React** and **Next.js** on the frontend. Experienced shipping end-to-end across product, design, architecture, and DevOps, including **LLM/RAG** systems, multi-agent pipelines, and CI/CD on Docker, AWS, and Heroku. Builds human-centered, modular, testable code and thrives in fast-iteration environments where one engineer owns a feature from spec to production.

CORE COMPETENCIES

- **Languages and Runtimes:** TypeScript, JavaScript (ESNext), Node.js, with working Python for AI/ML scripts.
- **Backend:** NestJS, Express-style Node.js services, REST APIs, GraphQL, JWT auth, RBAC, SSE, WebSockets.
- **Frontend:** React, Next.js, Vue.js (legacy maintenance), Redux, React Query, Zustand, Tailwind CSS, Radix UI.
- **Data:** PostgreSQL (with pgvector), MongoDB, Prisma, schema design, vector embeddings, hybrid search.
- **AI/LLMs:** OpenAI API, LangChain, Vercel AI SDK, Mastra multi-agent framework, RAG pipelines, agentic workflows, prompt engineering.
- **DevOps and CI/CD:** Docker / Docker Compose, GitHub Actions, AWS, Heroku, Vercel, Nx and pnpm monorepos, Playwright in CI.
- **Leadership:** Greenfield delivery, technical architecture reviews, mentorship of junior engineers and interns, async written communication.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension with **TypeScript**, owning the rebuild end-to-end across product, design, and engineering.
- Co-developed a full-stack **Next.js** chat widget powered by a **Retrieval-Augmented Generation (RAG)** system using OpenAI GPT models, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, letting users connect data sources, embed the widget on external sites, and query data conversationally.
- Implemented **JWT-based auth** and **role-based access control** on the frontend with HTTP-only cookies for access and refresh tokens, protected client-side routes, permission-gated UI, and seamless token refresh, collaborating with backend engineers on the auth contract.
- Introduced **Redux** for client state and **React Query** for server-state caching and request deduplication, refactored class components to functional components, and tuned the build pipeline for faster iteration.
- Optimized application performance through lazy loading, code splitting, image optimization, memoization, and CDN delivery, profiling with **Lighthouse** and Chrome DevTools to fix concrete bottlenecks.
- Integrated **Playwright** end-to-end tests in the **GitHub Actions** CI/CD pipeline, published **Allure Reports** on GitHub Pages, and pushed Slack notifications with deep links to test results on every push.
- Architected a reusable design system in **React** and **Radix UI** with **Tailwind CSS**, raising UI consistency, accessibility (WCAG/ARIA), and developer velocity across products.
- Maintained the existing **Vue.js** codebase in parallel with the React rebuild, fixing bugs and shipping features without regressions.
- Mentored two junior developers (interns), driving a culture of code review, testing, and craftsmanship.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of a greenfield **React** + **TypeScript** cloud console interface, achieving a **25% boost in user engagement** and partnering with architects on a scalable frontend that accelerated delivery by **30%**.
- Profiled render bottlenecks with **React Profiler** and **Lighthouse**, cut bundle size **30-40%** via rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster** initial load.
- Drove a **40% improvement** in runtime performance by reworking caching strategies across the React application.
- Introduced **Jest** and **Enzyme** testing frameworks plus **Storybook** for component documentation, leading to a **20% drop** in bug reports and clearer change management.
- Established team engineering standards via **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD.
- Mentored junior engineers on frontend best practices, fostering a culture of high-quality, maintainable code.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of transactions daily and boosting settlement rates by **80%**.
- Developed payment modules (MasterCard, VISA, Express-cash, E-commerce) with **React**, **Redux**, **RxJS** for data streaming, and **Axios** for asynchronous APIs.
- Built a real-time account reconciliation tool, lifting daily reconciliation by **60%** through new transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming in place of socket.io, improving update latency and UI responsiveness.
- Built dynamic data visualization dashboards with **react-chartjs**, **Redux Toolkit**, and **Ant Design** for role and permission management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Translated business requirements into efficient **MongoDB** schemas, partnering with the business team to keep the data model aligned with product goals.
- Developed **NodeJS** RESTful APIs for new product features, focused on testable, maintainable, well-documented code.
- Converted user journeys and Figma designs into reusable **React** components for a consistent UI.
- Engineered a CI/CD pipeline for automated API deployment to **Heroku** testing environments and production servers on **AWS**.
- Mentored two interns through frontend and backend projects with structured technical guidance.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** schemas and implemented server-side logic in **NodeJS** for multiple production applications.
- Built **SureEdu**, a school management portal, in a team of 5: staff payroll and attendance, student records and exams, guardian portals.
- Owned full-stack integration: connected the JavaScript frontend to NodeJS services, built and tested REST endpoints, authored use cases, and translated them into database schemas.
- Wrote comprehensive documentation for each application detailing functions, capabilities, and feature behavior.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. Multi-agent LLM pipeline analyzing 700+ budget documents (708K+ vector embeddings) across 37 states.

- Architected the entire platform as an **Nx monorepo** of **6 applications**: **NestJS** API backend, **NestJS** document ingestion pipeline, **Next.js** citizen-facing web app, **Next.js** admin dashboard, **Next.js** marketing landing page, and a Remotion video service, all orchestrated with **pnpm** workspaces.
- Designed and shipped a **multi-agent AI system** on the **Mastra** framework with 5 specialized agents (Router, Budget, Corruption, GovSpend, Impact) using three-stage intent classification and domain-specific RAG analysis.
- Built a hybrid **RAG** pipeline combining **pgvector** similarity search with BM25 full-text search merged via Reciprocal Rank Fusion, with Cohere reranking and Voyage AI embeddings (1024-dim, 708K+ chunks).
- Configured CI/CD with **GitHub Actions**, **Playwright** E2E tests across 4 projects (web auth, dashboard auth, desktop, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and 30s graceful shutdown.
- Integrated **Langfuse** for LLM observability, **Infisical** for secrets management, and **Traefik** for reverse proxy routing in production.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*